CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 4: **Electricity: Magnetic and Heating Effects** | Quick revision, one page

Electric circuit

- A circuit has a source (cell/battery), conducting wires, a switch and the device (bulb, heater).
- Current flows only in a closed circuit; conventional current goes from + to - outside the cell.
- Conductors (metals, graphite) allow current; insulators (plastic, rubber, dry wood) do not.
- Unit of electric current: ampere (A).

Magnetic effect of current

- Oersted: a current-carrying wire deflects a magnetic compass needle.
- Reversing the current reverses the deflection.
- Electromagnet: insulated wire coiled on a soft iron core; it is magnetic only while current flows.

Strength of an electromagnet

- Increases with: more turns, more current (more cells), a soft iron core.
- Uses: lifting scrap iron, electric bell, relays, cranes.

Heating effect of current

- A current-carrying conductor becomes hot; heat increases with current, resistance and time.
- Heating elements (heater, iron, toaster): nichrome, high resistance and high melting point.
- Bulb filament: tungsten, very high melting point.
- Fuse: thin low-melting wire; rating just above the normal current; MCB trips and can be reset.

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